University of British Columbia, Vancouver
Department of Computer Science

CPSC 304 Project Cover Page

Milestone #: 1

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Group Number: 4

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

CPSC304 Project Milestone 1

1 Project Description

The domain of our application is art collection management. Our application will serve art dealerships/management companies who are responsible for large private art collections. Our users are employed by clients with valuable works of art that may be loaned to museums. Each piece of art may have specific security and maintenance requirements. The art dealership must also maintain a record of each art work's current museum and the contact information for the conservator. Our application will offer a convenient way for art dealerships to store and access all of this information.

The applications database will model the most important information for these collection managers. We will represent each client and all of their art. Each piece of artwork will be related to its medium, artist and the current museum. We will also represent the required security and maintenance for a piece of art and the conservator who is responsible for said maintenance. For example, a client with a large art collection passes away and leaves all his art to their family. The art dealer who managed this client's collection may then receive a request from the next of kin for a dossier summarizing the client's portfolio. The dealer can use this database to quickly obtain a comprehensive list of the client's art (including its artist and medium), where it is housed, who the conservator is and how to contact them, as well as the security and maintenance requirements. The client's family can use all of this information to decide if they will sell the artwork or continue to maintain the collection.

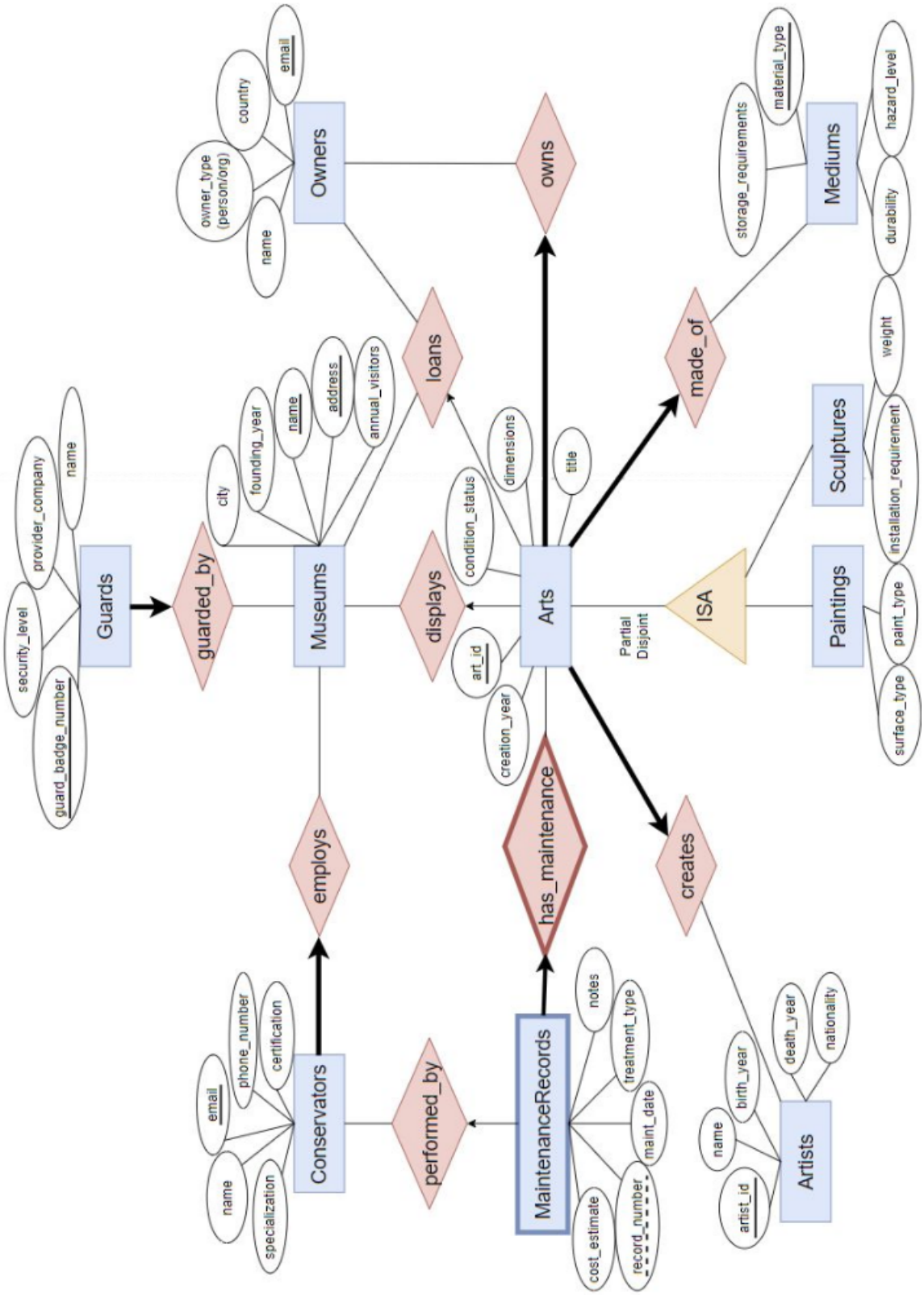
2 Database Specification

1. As an owner of the fine art portfolio manager service, I should be able to count the number of fine art in our system so I can advertise to fine art portfolio managers.

2. As an fine art portfolio manager, I should be able to query for an owners loans of art to museums so I can describe the loans to an owner.

3. As an fine art portfolio manager, I should be able to query for fine art in the system that has the same artist so I can pair up owners of Van Gogh art.

4. As an fine art portfolio manager, I should be able to calculate the insurance premium for an art as a function of maintainance records, conservators, and guards at the museum that displays the art so I can put a price on managing an art of an owner.


Guards
security_level
provider_company
name
guard_badge_number
guarded_by
Museums
city
founding_year
name
address
annual_visitors
loans
Owners
owner_type (person/org)
country
email
name
owns
displays
condition_status
dimensions
title
art_id
creation_year
Arts
Partial
Disjoint
ISA
made_of
storage_requirements
material_type
Mediums
hazard_level
durability
weight
Sculptures
installation_requirement
Paintings
paint_type
surface_type
employs
has_maintenance
creates
Conservators
email
phone_number
certification
name
specialization
performed_by
MaintenanceRecords
notes
treatment_type
maint_date
cost_estimate
record_number
Artists
birth_year
death_year
nationality
name
artist_id

3 ER Diagram

We used two artificial keys in our ER diagram: art_id and artist_id. The use of artificial keys for these entities is justified by likely duplication in the entities' other attributes. For art, there may be more than one piece with the same title and same year (potentially with the same dimensions and condition). Similarly, for artist it is possible for two artists to share the same name, birth year and nationality (e.g. John Smith, 1995, n/a, USA). Using artificial keys is the best way to reliably identify each work of art and artist.

4 AI Acknowledgement

No member of our team used any AI tools to complete this milestone.